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Array substrate, display panel, and alignment method of liquid crystal molecules

Abstract

An array substrate includes a first substrate, data lines, a pixel electrode layer, first transparent common electrode lines, and second transparent common electrode lines disposed on the first substrate. The data lines extend in a first direction and are arranged at intervals in a second direction intersecting the first direction. The pixel electrode layer includes a plurality of pixel electrodes. Both the first transparent common electrode lines and the second transparent common electrode lines extend in the first direction and are disposed insulated from the pixel electrode layer, and the second transparent common electrode lines are disposed disconnected from the first transparent common electrode lines. An orthographic projection of a gap between a first and a second transparent common electrode line adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of at least one pixel electrode on the first substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the benefit of priority of Chinese Patent Application No.

202311267552.4, filed on Sep. 26, 2023, the contents of which are incorporated by reference as if fully set forth herein in their entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the field of display technologies, and more particularly, to an array substrate, a display panel, and an alignment method of liquid crystal molecules.

BACKGROUND

(3) In recent years, the liquid crystal display (LCD) industry has developed rapidly, and great breakthroughs have been made in key technologies such as large size, curved surface, and ultra-high resolution. Although new display technologies, such as miniature light-emitting diodes (Mini LEDs), organic light emitting diode (OLEDs), and quantum-dot light emitting diodes (QLEDs), have begun to appear in the market, they have been slowly developed due to the dual limitations of raw materials and precision of equipment. Thus, in the coming years, LCD still occupies an absolute dominant position in the display industry due to mature technology, complete material system, and low cost. The improvement of transmittance is one of the effective ways for LCD to achieve “low power consumption”.

(4) Therefore, how to improve the transmittance of LCD is a technical problem to be solved.

SUMMARY

(5) In a first aspect, the present disclosure provides an array substrate including: a first substrate; a plurality of data lines extending in a first direction and arranged at intervals in a second direction, disposed on the first substrate, the first direction intersecting the second direction; a pixel electrode layer disposed on the first substrate and including a plurality of pixel electrodes; first transparent common electrode lines extending in the first direction, disposed on the first substrate and insulated from the pixel electrode layer; and second transparent common electrode lines extending in the first direction, disposed on the first substrate and insulated from the pixel electrode layer, and disposed disconnected from the first transparent common electrode lines; where an orthographic projection of a gap between one of the first transparent common electrode lines and one of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of at least one of the pixel electrodes on the first substrate.

(6) In a second aspect, the present disclosure provides a display panel including an array substrate, an opposing substrate, and a liquid crystal layer, the opposing substrate is disposed opposite to the array substrate, and the liquid crystal layer is disposed between the array substrate and the opposing substrate; the array substrate includes: a first substrate; a plurality of data lines extending in a first direction and arranged at intervals in a second direction, disposed on the first substrate, the first direction intersecting the second direction; a pixel electrode layer disposed on the first substrate and including a plurality of pixel electrodes; first transparent common electrode lines extending in the first direction, disposed on the first substrate and insulated from the pixel electrode layer; and second transparent common electrode lines extending in the first direction, disposed on the first substrate and insulated from the pixel electrode layer, and disposed disconnected from the first transparent common electrode lines; where an orthographic projection of a gap between one of the first transparent common electrode lines and one of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of at least one of the pixel electrodes on the first substrate.

(7) In a third aspect, the present disclosure provides an alignment method of liquid crystal molecules, applied to a display panel, where the display panel includes an array substrate and an opposing substrate disposed opposite each other, and a liquid crystal layer disposed between the array substrate and the opposing substrate and including a plurality of liquid crystal molecules; the array substrate includes a first substrate and a pixel electrode layer disposed on the first substrate, first transparent common electrode lines extending in a first direction, second transparent common electrode lines extending in the first direction, and a plurality of data lines extending in a first

direction and arranged at intervals in a second direction; the pixel electrode layer including a plurality of pixel electrodes, the first transparent common electrode lines and the second transparent common electrode lines disposed disconnected from each other and insulated from the pixel electrode layer; an orthographic projection of a gap between one of the first transparent common electrode lines and one of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of at least one of the pixel electrodes on the first substrate, the first direction intersecting the second direction; the method including: applying a first preset alignment voltage to the first transparent common electrode lines; applying a second preset alignment voltage to the second transparent common electrode lines, the second preset alignment voltage being different from the first preset alignment voltage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic cross-sectional view of a display device according to some embodiments of the present disclosure;
- (2) FIG. 2 is a schematic plan view of a transparent conductive layer according to some embodiments of the present disclosure;
- (3) FIG. 3 is a schematic plan view of a pixel electrode layer, a first metal layer, a second metal layer, a semiconductor layer, and a transparent conductive layer according to some embodiments of the present disclosure;
- (4) FIG. 4 is a schematic plan view of a pixel electrode layer, a first metal layer, a second metal layer, a semiconductor layer, and a transparent conductive layer according to some other embodiments of the present disclosure;
- (5) FIG. 5 is a schematic plan view of a pixel electrode layer, a first metal layer, a second metal layer, a semiconductor layer, and a transparent conductive layer according to still some other embodiments of the present disclosure;
- (6) FIG. 6 is a schematic flow diagram of an alignment method of liquid crystal molecules of a display panel of a display device according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

- (7) The technical solutions in the embodiments of the present disclosure will be clearly and completely described below with reference to the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only some of the embodiments of the present disclosure, but not all of the embodiments. Based on the embodiments in the present disclosure, all other embodiments obtained by those skilled in the art without making creative efforts fall within the scope of protection of the present disclosure.
- (8) Referring to FIGS. 1 to 3, FIG. 1 is a schematic cross-sectional view of a display device according to some embodiments of the present disclosure, FIG. 2 is a schematic plan view of a transparent conductive layer according to some embodiments of the present disclosure, and FIG. 3 is a schematic plan view of a pixel electrode layer, a first metal layer, a second metal layer, a semiconductor layer, and a transparent conductive layer according to some embodiments of the present disclosure.
- (9) As shown in FIG. 1, the display device **300** includes a display panel **100** and a backlight module **200** located on a back side of a light-emitting side of the display panel **100**.
- (10) The display panel **100** includes an array substrate **10**, an opposing substrate **20**, and a liquid crystal layer **30**. The array substrate **10** is disposed opposite to the opposing substrate **20**. A liquid crystal layer **30** is disposed between the array substrate **10** and the opposing substrate **20** and includes a plurality of liquid crystal molecules **301**.

(11) The array substrate **10** includes a first substrate **101**, a first metal layer **102**, a semiconductor layer **103**, a second metal layer **104**, a transparent conductive layer **105**, and a pixel electrode layer **106**. The transparent conductive layer **105** is provided insulated from the pixel electrode layer **106**. As shown in FIG. 2, the array substrate **10** includes an array region **10a** and a peripheral region **10b** located at a periphery of the array region **10a**. The array area **10a** corresponds to a display area of the display panel **100**. The peripheral area **10b** corresponds to a non-display area of the display panel **100**. The opposing substrate **20** includes a second substrate **201** and a common electrode layer **202**. The common electrode layer **202** is disposed on a surface of the second substrate **201** adjacent to the array substrate **10**.

(12) The first substrate **101** is disposed opposite to the second substrate **201**. The first substrate **101** and the second substrate **201** are glass substrates, but are not limited thereto.

(13) As shown in FIGS. 1 and 3, the first metal layer **102** is disposed on a surface of the first substrate **101** adjacent to the second substrate **201**. The first metal layer **102** includes a gate **1021** and a scan line **1022**. The gate **1021** is connected to a scan line **1022**, which extends in a second direction y.

(14) The semiconductor layer **103** is located on a side of the first metal layer **102** away from the first substrate **101**. The semiconductor layer **103** includes an active layer **1031** that overlaps with the gate **1021**. A first insulating layer **1071** is provided between the semiconductor layer **103** and the first metal layer **102**. It will be appreciated that the semiconductor layer **103** may also be disposed between the first metal layer **102** and the first substrate **101**.

(15) As shown in FIGS. 1 and 3, the second metal layer **104** is located on a side of the semiconductor layer **103** away from the first substrate **101**. A second insulating layer **1072** is provided between the second metal layer **104** and the semiconductor layer **103**. The second metal layer **104** includes a source **1041** and a drain **1042** connected to the active layer **1031**. The second metal layer **104** also includes a plurality of data lines **1043**. The plurality of data lines **1043** extend in a first direction x and are arranged at intervals in the second direction y. The first direction x intersects the second direction y.

(16) Specifically, the first direction x is perpendicular to the second direction y, but is not limited thereto. The included angle between the first direction x and the second direction y may also be an obtuse angle or an acute angle.

(17) It should be noted that the source **1041**, the drain **1042**, the active layer **1031**, and the gate **1021** constitute a thin-film transistor T. The source **1041** of the thin-film transistor T is connected to a data line **1043**.

(18) As shown in FIGS. 1, 2, and 3, the pixel electrode layer **106** is also disposed on the first substrate **101**. The pixel electrode layer **106** includes a plurality of pixel electrodes **1061**. The plurality of pixel electrodes **1061** are arranged in an array in the array region **10a** in the first direction x and the second direction y. Each pixel electrode **1061** is connected to the drain **1042** of the thin-film transistor T. The material of the pixel electrode layer **106** includes at least one of indium tin oxide and indium zinc oxide.

(19) As shown in FIG. 3, each pixel electrode **1061** includes a first trunk electrode **1062** extending in the first direction x, a second trunk electrode **1063** extending in the second direction y, a plurality of branch electrodes **1064**, and a plurality of apertures **1065**. The first trunk electrode **1062** is connected to a connection portion **1066**. The connection portion **1066** is connected to the drain **1042** of the thin-film transistor T through a via hole, so that the first trunk electrode **1062** is connected to the drain **1042** of the thin-film transistor T.

(20) One end of the second trunk electrode **1063** is connected between both ends of the first trunk electrode **1062**, and the pixel electrode **1061** is in turn divided into two domain regions. The plurality of branch electrodes **1064** are located on both sides of the second trunk electrode **1063** in the first direction x and are connected to the first trunk electrode **1062** and/or the second trunk electrode **1063**, respectively. An aperture **1065** is provided between two adjacent branch electrodes

1064 in one domain region. Therefore, the pixel electrode **1061** is a two-domain pixel electrode.

(21) It should be noted that compared with four-domain pixel electrodes in some related art, the two-domain pixel electrode of the present disclosure has fewer trunk electrodes, which is more advantageous for improving the aperture ratio of the pixel electrode. As such, the light transmission of the display panel **100** is higher, thereby improving the display effect of the display device **300**. In addition, the pixel electrode **1061** shown in FIG. **3** includes only one first trunk electrode **1062**, which further reduces the number of trunk electrodes and is more advantageous for the improvement of the aperture ratio of the pixel electrode. The light transmittance of the display panel **100** is higher, and the display effect of the display device **300** is further improved.

(22) In the case that two-domain pixel electrodes are used to increase the aperture ratio, the pixel electrodes **1061** are of asymmetric structures in the second direction y. Here, the reason for the asymmetry of a pixel electrode **1061** in the second direction y shown in FIG. **3** includes that the pixel electrode **1061** includes a first trunk electrode **1062** extending in the first direction x, and does not include an electrode symmetrical to the first trunk electrode **1062**, and that the plurality of branch electrodes **1064** are not symmetrical in the second direction y either.

(23) As shown in FIG. **1**, the transparent conductive layer **105** is disposed on the first substrate **101**. The transparent conductive layer **105** is located on a side, away from the first substrate **101**, of a film layer on which the plurality of data lines **1043** are located. The transparent conductive layer **105** is located between the film layer on which the plurality of data lines **1043** are located and the pixel electrode layer **106**. As such, at least a portion of the transparent conductive layer **105** overlaps with the data line **1043**. At least the portion of the transparent conductive layer **105** replaces a shielding electrode in the related art that locates on a same layer as the pixel electrode **1061** does. At least the portion of the transparent conductive layer **105** functions as the shielding electrode, facilitating an increase in the layout area of the pixel electrode **1061**, and improving the light transmittance of the display panel **100**.

(24) Specifically, as shown in FIG. **1**, the transparent conductive layer **105** is located on a side of the second metal layer **104** away from the first substrate **101**. The pixel electrode layer **106** is located on a side of the transparent conductive layer **105** away from the first substrate **101**. A third insulating layer **1073** is provided between the transparent conductive layer **105** and the second metal layer **104**. A fourth insulation layer **1074** is provided between the pixel electrode layer **106** and the transparent conductive layer **105**.

(25) In some embodiments, the first insulating layer **1071**, the second insulating layer **1072**, and the third insulating layer **1073** are all inorganic insulating layers. The fourth insulating layer **1074** is an organic insulating layer, and the organic insulating layer plays a flattening role. Materials of the inorganic insulating layers are selected from one or more of silicon oxide, silicon nitride, and silicon oxynitride. Materials of the organic insulating layer include one or more of polyimides and polyacrylates.

(26) In some embodiments, as shown in FIG. **1**, the array substrate **10** further includes a color resistance layer **108**. The color resistance layer **108** is located between the second metal layer **104** where the plurality of data lines **1043** are located and the transparent conductive layer **105**. Here, the color resistance layer **108** includes a plurality of color resistances of different colors, such as a red color resistance, a blue color resistance, and a green color resistance.

(27) Specifically, a third insulating layer **1073** is provided between the color resistance layer **108** and the second metal layer **104**. The transparent conductive layer **105** is located on a surface of the color resistance layer **108** away from the first substrate **101**.

(28) It should be noted that the color resistance layer **108** is provided between the transparent conductive layer **105** and the second metal layer **104** on which the plurality of data lines **1043** are located. As such, parasitic capacitances between the transparent conductive layer **105** and the plurality of data lines **1043** can be reduced, and the effect of the parasitic capacitances on data signals transmitted by the data lines **1043** can be reduced.

(29) In some other embodiments, the transparent conductive layer **105** may also be located between the second metal layer **104** on which the plurality of data lines **1043** are located and the first substrate **101**. The pixel electrode layer **106** is located on the side of the second metal layer **104** away from the first substrate **101**, that is, the transparent conductive layer **105** is located below the second metal layer **104**. In some other embodiments, the transparent conductive layer **105** may also be located between the first metal layer **102** on which a plurality of the scan lines **1022** are located and the first substrate **101**, i.e., the transparent conductive layer **105** is located below the first metal layer **102**.

(30) The transparent conductive layer **105** includes a plurality of first transparent common electrode lines **1051** arranged at intervals in the second direction y and a plurality of second transparent common electrode lines **1052** arranged at intervals in the second direction y. The plurality of first transparent common electrode lines **1051** and the plurality of second transparent common electrode lines **1052** are both located in the array region **10a** and extend in the first direction x. That is, the extension direction of the first transparent common electrode lines **1051** and the second transparent common electrode lines **1052** is the same as the extension direction of the data lines **1043**.

(31) A first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y are disconnected and arranged at intervals. Therefore, the first transparent common electrode lines **1051** and the second transparent common electrode lines **1052** are provided in a same layer to simplify the manufacturing process of both. In addition, both the first transparent common electrode lines **1051** and the second transparent common electrode lines **1052** are transparent, thereby improving the light transmittance of the display panel. Materials of the transparent conductive layer **105** includes at least one of indium tin oxide and indium zinc oxide.

(32) In some other embodiments, the first transparent common electrode lines **1051** and the second transparent common electrode lines **1052** may be located at different transparent conductive layers, respectively. As such, flexibility in manufacturing the first transparent common electrode lines **1051** and the second transparent common electrode lines **1052** is improved.

(33) It should be noted that the first transparent common electrode lines **1051** and the second transparent common electrode lines **1052** being provided in a same layer means that the first transparent common electrode lines **1051** and the second transparent common electrode lines **1052** are simultaneously prepared and obtained by patterning a same film layer. The first transparent common electrode lines **1051** and the second transparent common electrode lines **1052** being located on different transparent conductive layers respectively means that the first transparent common electrode lines **1051** and the second transparent common electrode lines **1052** are prepared and obtained by patterning two different film layers.

(34) Specifically, the plurality of first transparent common electrode lines **1051** and the plurality of second transparent common electrode lines **1052** are alternately arranged one-to-one in the second direction y.

(35) In some embodiments, in the case where the transparent conductive layer **105** is located between the second metal layer **104** on which the plurality of data lines **1043** are located and the pixel electrode layer **106**, an orthographic projection of at least one of the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** overlaps with an orthographic projection of at least one of the data lines **1043** on the first substrate **101**. Thus, at least one of the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other in the second direction y may act as a shield for signals transmitted by the at least one of the data lines **1043**. At least one of the first transparent common electrode line **1051** and the second transparent common electrode line **1052** replaces a shielding electrode in the related art that locates on a same layer as the pixel electrode **1061** does, thereby increasing the layout area of the pixel

electrode **1061** and further improving the light transmittance of the display panel **100**.

(36) Specifically, as shown in FIG. 3, orthographic projections of a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** overlap with orthographic projections of two adjacent ones of the data lines **1043** on the first substrate **101**, respectively. With this arrangement, a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y both function as a shielding electrode, further increasing the layout area of the pixel electrode **1061**, and further improving the light transmittance of the display panel **100**.

(37) In some other embodiments, an orthographic projection of either of the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other on the first substrate **101** may overlap with an orthographic projection of one of the data lines **1043** on the first substrate **101**.

(38) In some embodiments, an orthographic projection of at least one of a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in a second direction y on the first substrate **101** overlaps with an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101**. By this design, except that a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y may function as a shielding electrode, at least one of the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other in the second direction y may form a storage capacitor with the at least one of the pixel electrodes **1061** to store display data.

(39) In some embodiments, as shown in FIG. 3, an orthographic projection of one of a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** may overlap with an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101**, and an orthographic projection of another one of the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** may not overlap with an orthographic projection of the pixel electrode **1061** on the first substrate **101**.

(40) In some other embodiments, the orthographic projections of a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** may not overlap with orthographic projections of the pixel electrodes **1061** on the first substrate **101**, but are provided adjacent to an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101**.

(41) In some other embodiments, the orthographic projections of a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** may both overlap with an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101**.

(42) In some embodiments, the orthographic projections of a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** both at least overlap with an orthographic projection of a trunk electrode of the pixel electrode **1061** on the first substrate **101**.

(43) For example, for a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other, the orthographic projection of the first transparent common electrode line **1051** on the first substrate **101** overlaps with an orthographic projection of a first trunk electrode **1062** of at least one of the pixel electrodes **1061** on the first substrate **101**, and the orthographic projection of the second transparent common electrode line **1052** on the first substrate **101** at least overlaps with an orthographic projection of a second trunk electrode **1063** of the pixel electrode **1061** on the first substrate **101**. With this arrangement, when

the storage capacitor is formed, the liquid crystal molecules above the trunk electrodes are better controlled to be tilted towards outside of the aperture region, thereby alleviating dark fringes around the trunk electrodes.

(44) In some embodiments, in the case where both orthographic projections of a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** overlap with an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101**, areas of the orthographic projections of the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** differ from that of the orthographic projection of one pixel electrode **1061** on the first substrate **101**. As such, capacitance values of the storage capacitors formed by the first transparent common electrode line **1051** and the second transparent common electrode line **1052** with a same pixel electrode **1061** are different so as to accommodate different capacitance requirements.

(45) In some other embodiments, in the case where orthographic projections of a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** overlap with orthographic projections of the pixel electrodes **1061** on the first substrate **101**, areas of the orthographic projections of the first transparent common electrode line **1051** and the second transparent common electrode line **1052** on the first substrate **101** may be same as that of an orthographic projection of a same pixel electrode **1061** on the first substrate **101**. As such, capacitance values of the storage capacitors formed by the first transparent common electrode line **1051** and the second transparent common electrode line **1052** with the same pixel electrode **1061** are the same.

(46) Orthographic projections of a second transparent common electrode line **1052** and a first transparent common electrode line **1051** adjacent to each other in the second direction y on the first substrate **101** are provided adjacent to an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101**. An orthographic projection of a gap between the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** overlaps with the orthographic projection of the at least one of the pixel electrodes **1061** on the first substrate **101**. With this arrangement, during an alignment process of a plurality of liquid crystal molecules **301** of the liquid crystal layer **30**, a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y constitute a pair of transparent common electrode lines. A voltage applied to the pair of transparent common electrode lines cooperates with a voltage applied to a pixel electrode **1061** to align a plurality of liquid crystal molecules **301** controlled by the pixel electrode **1061**. Voltages applied to multiple pairs of transparent common electrode lines cooperate with voltages applied to a number of the pixel electrodes **1061** to align a plurality of liquid crystal molecules **301** controlled by the number of the pixel electrodes **1061**.

(47) In addition, during operation of the display device **300**, the plurality of first transparent common electrode lines **1051** and the plurality of second transparent common electrode lines **1052** are applied with same constant voltages, respectively. Thus, the plurality of first transparent common electrode lines **1051** and the plurality of second transparent common electrode lines **1052** have different roles during alignment of the liquid crystal molecules **301** and during display of the display device **300**.

(48) It should be noted that in the case where an orthographic projection of a gap between a second transparent common electrode line **1052** and a first transparent common electrode line **1051** adjacent to each other in the second direction y on the first substrate **101** overlaps with an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101**, at least three designs (A), (B) and (C) may be included. For design (A), orthographic projections of a second transparent common electrode line **1052** and a first transparent common electrode line **1051**

adjacent to each other in the second direction y on the first substrate **101** may both overlap with an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101**. For design (B), an orthographic projection of one of the second transparent common electrode lines **1052** and the first transparent common electrode lines **1051** adjacent to each other in the second direction y on the first substrate **101** may overlap with an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101**, and an orthographic projection of another one of the second transparent common electrode lines **1052** and the first transparent common electrode lines **1051** adjacent to each other in the second direction y on the first substrate **101** may not overlap with the orthographic projection of the pixel electrode **1061** on the first substrate **101**. For design (C), orthographic projections of a second transparent common electrode line **1052** and a first transparent common electrode line **1051** adjacent to each other in the second direction y on the first substrate **101** may be positioned on opposite sides of an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101** in the second direction y, respectively.

(49) FIG. 3 is only a schematic diagram illustrating an orthographic projection of a gap between a second transparent common electrode line **1052** and a first transparent common electrode line **1051** adjacent to each other in the second direction y on the first substrate **101** overlapping with an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101**. For the plurality of pixel electrodes **1061** of the array region **10a**, orthographic projections of a second transparent common electrode line **1052** and a first transparent common electrode line **1051** adjacent to each other in the second direction y on the first substrate **101** overlap with orthographic projections of a number of pixel electrodes **1061** of a row of pixel electrodes **1061** disposed side-by-side in the first direction x on the first substrate **101**.

(50) In some embodiments, as shown in FIG. 3, the orthographic projection of the first transparent common electrode line **1051** on the first substrate **101** is provided adjacent to the orthographic projection of the first trunk electrode **1062** on the first substrate **101**. The width of the first transparent common electrode line **1051** is larger than the width of the second transparent common electrode line **1052**. The width of the first transparent common electrode line **1051** and the width of the second transparent common electrode line **1052** are both larger than the width of the data line **1043**. With this arrangement, the first transparent common electrode line **1051** may cover the adjacent data line **1043** and the first trunk electrode **1062**.

(51) In some embodiments, an orthographic projection of a first transparent common electrode line **1051** on the first substrate **101** overlaps with orthographic projections of a first trunk electrode **1062** of a row of pixel electrodes **1061** disposed side-by-side in the first direction x and a data line **1043** on the first substrate **101**. An orthographic projection of a second transparent common electrode line **1052** on the first substrate **101** overlaps with an orthographic projection of another data line **1043** on the first substrate **101**. Moreover, an orthographic projection of a gap between a second transparent common electrode line **1052** and a first transparent common electrode line **1051** adjacent to each other in the second direction y on the first substrate **101** overlaps with orthographic projections of a plurality of branch electrodes **1064** of the pixel electrode **1061** on the first substrate **101**. With this arrangement, the first transparent common electrode line **1051** and the second transparent common electrode line **1052** both function as a shielding electrode, and a storage capacitor is formed between the first transparent common electrode line **1051** and the pixel electrode **1061**.

(52) In the case where the pixel electrode **1061** is of an asymmetric structure in the second direction y, if the first transparent common electrode line **1051** and the second transparent common electrode line **1052** of the present disclosure are not provided, the light has a low transmittance in the vicinity of the first trunk electrode **1062** of the pixel electrode **1061**, resulting in dark fringes in the vicinity of the first trunk electrode **1062** in the second direction y.

(53) In the present disclosure, an electric field corresponding to a voltage difference between the adjacent pixel electrode **1061** and the first transparent common electrode line **1051** is

independently controlled. An electric field corresponding to a voltage difference between the adjacent pixel electrode **1061** and the second transparent common electrode line **1052** is independently controlled. The pretilt angle(s) of the liquid crystal molecules **301** under an action of the electric field formed between the first transparent common electrode line **1051** and the pixel electrode **1061** and the pretilt angle(s) of the liquid crystal molecules **301** under an action of the electric field formed between the second transparent common electrode line **1052** and the pixel electrode **1061** may both be independently controlled. Providing conditions for the liquid crystal molecules **301** to tilt towards an area outside of the aperture region of the pixel electrode **1061** in the second direction y is advantageous for ameliorating the problem of dark fringes of the pixel electrode **1061** in the second direction y, which in turn improves the transmittance of the display panel **100** with respect to the backlight emitted by the backlight module **200**, and improves the display effect of the display device **300**. In particular, in the case where the orthographic projection of the first transparent common electrode line **1051** on the first substrate **101** overlaps with the orthographic projection of the first trunk electrode **1062** of the row of pixel electrodes **1061** disposed side-by-side in the first direction x on the first substrate **101**, the voltage applied to the first transparent common electrode line **1051** is controlled so that the liquid crystal molecules **301** above the first trunk electrode **1062** may be tilted towards outside of the aperture, thereby ameliorating the problem that the light has a low transmittance in the vicinity of the first trunk electrode **1062** of the pixel electrode **1061** and dark fringes in turn occur in the second direction y. (54) It should be noted that in some related art, when an entire transparent conductive layer is used with a pixel electrode layer and a common electrode to align the liquid crystals, only one voltage can be applied to the entire transparent conductive layer, so that the localized dark fringes occurring in the display of the display panel cannot be improved. In addition, compared with the related art in which common electrode lines are metal lines etc., which will reduce the transmittance of the display panel, the present disclosure employs the first transparent common electrode lines **1051** and the second trunk common electrode line **1054** for the alignment of the liquid crystal molecules **301**, which may also improve the transmittance of the display panel **100** with respect to the backlight emitted by the backlight module **200**.

(55) In addition, in the related art, the size of the pixel electrodes in the first direction x is larger, and correspondingly, the area of the dark fringes of the pixel electrodes in the second direction y is larger. In order to solve the problem that the area of the dark fringes of the pixel electrodes in the second direction y is larger, the present disclosure adds a first transparent common electrode line **1051** and a second transparent common electrode line **1051** adjacent to each other and arranged at intervals in the second direction y for each pixel electrode. An orthographic projection of a gap between a second transparent common electrode line **1052** and a first transparent common electrode line **1051** adjacent to each other in the second direction y on the first substrate **101** overlaps with an orthographic projection of at least one of the pixel electrodes **1061** on the first substrate **101**, which distinctly ameliorates the problem of dark fringes of pixel electrodes in the second direction y, and in turn significantly improves the light transmittance of the display panel **100**.

(56) As can be seen from the above, a second transparent common electrode line **1052** and a first transparent common electrode line **1051** adjacent to each other in the second direction y may be used not only as a shielding electrode of the data line **1043**, but also as an electrode plate of the storage capacitor, and may as well be used to ameliorate the problem of dark fringes in the vicinity of the trunk electrode of the pixel electrode extending in the first direction x. With this arrangement, the shielding electrode and the electrode plate in the related art may be omitted, and the light transmittance of the display panel may be significantly improved.

(57) It will be appreciated that when the pixel electrode **1061** includes a lateral trunk electrode extending in the second direction y and located at the outermost side of the pixel electrode **1061**, the orthographic projection of the lateral trunk electrode on the first substrate **101** may also overlap

with the orthographic projection of the third common electrode line (not shown) on the first substrate **101**. As such, the problem of dark fringes in the vicinity of the lateral trunk electrode is ameliorated.

(58) As shown in FIG. 2, the array substrate **10** further includes a first trunk common electrode line **1053** extending in the second direction y and a second trunk common electrode line **1054** extending in the second direction y. The first trunk common electrode line **1053** is located in the peripheral region **10b** and is connected to the plurality of first transparent common electrode lines **1051**. The second trunk common electrode line **1054** is located in the peripheral region **10b** and is disposed disconnected from the first trunk common electrode line **1053** and connected to the plurality of second transparent common electrode lines **1052**. The first trunk common electrode line **1053** and the second trunk common electrode line **1054** are located on opposite sides of the array region **10a** in the first direction x. As such, the first trunk common electrode line **1053** and the second trunk common electrode line **1054** are connected to the first transparent common electrode line **1051** and the second transparent common electrode line **1052**, respectively, and the risk of the first trunk common electrode line **1053** and the second trunk common electrode line **1054** being short-circuited is reduced.

(59) Specifically, the transparent conductive layer **105** further includes a first trunk common electrode line **1053** and a second trunk common electrode line **1054**. With this arrangement, the manufacturing process of the array substrate is simplified and the manufacturing process of the display panel is simplified.

(60) It will be appreciated that at least a portion of the first trunk common electrode line **1053** and at least a portion of the second trunk common electrode line **1054** may also be disposed in the array region **10a**. With this arrangement, it is advantageous for the display panel to realize the structure of a narrow bezel.

(61) In the present disclosure, different alignment voltages may be applied to the first trunk common electrode line **1053** and the second trunk common electrode line **1054**, respectively, during alignment of the liquid crystal molecules **301**. During operation of the display device **300**, same constant voltages are applied to the first trunk common electrode line **1053** and the second trunk common electrode line **1054**, respectively, so as to apply the same constant voltage to the plurality of first transparent common electrode lines **1051** and the plurality of second transparent common electrode lines **1052**, respectively. Referring to FIG. 4, there is shown a schematic plan view of a pixel electrode layer, a first metal layer, a second metal layer, a semiconductor layer, and a transparent conductive layer in some other embodiments of the present disclosure. In some other embodiments, the orthographic projection of at least one of the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** overlaps with the orthographic projection of the branch electrode **1064** on the first substrate **101**. With this arrangement, the overlapping area of at least one of the first transparent common electrode line **1051** and the second transparent common electrode line **1052** with the pixel electrode **1061** is increased, the capacitance value of the storage capacitor is increased, and the display problem caused by electric leakage of the storage capacitor is ameliorated.

(62) In some embodiments, an area where an orthographic projection of a first transparent common electrode line **1051** on the first substrate **101** overlaps with an orthographic projection of the first trunk electrode **1062** on the first substrate **101** is larger than an area where an orthographic projection of a second transparent common electrode line **1052** on the first substrate **101** overlaps with an orthographic projection of the first trunk electrode **1062** on the first substrate **101**. With this arrangement, the problem of dark fringes in the vicinity of the first trunk electrode **1062** is ameliorated through the first transparent common electrode line **1051**.

(63) In some embodiments, the area of a second transparent common electrode line **1052** is greater than the area of a first transparent common electrode line **1051**. With this arrangement, it is

advantageous that the capacitance value of the storage capacitor formed by a second transparent common electrode line **1052** and a pixel electrode is greater than the capacitance value of the storage capacitor formed by a first transparent common electrode line **1051** and a pixel electrode. (64) In some embodiments, the area where the orthographic projection of a first transparent common electrode line **1051** on the first substrate **101** overlaps with the orthographic projection of a plurality of branch electrodes **1064** of a pixel electrode **1061** on the first substrate **101** is smaller than the area where the orthographic projection of a second transparent common electrode line **1052** on the first substrate **101** overlaps with the orthographic projection of a plurality of branch electrodes **1064** of a pixel electrode **1061** on the first substrate **101**. With this arrangement, the capacitance value of the storage capacitor formed between the second transparent common electrode line **1052** and the pixel electrode **1061** is larger, thereby ameliorating the display problem caused by electric leakage of the storage capacitor.

(65) In a specific embodiment, the orthographic projection of a first transparent common electrode line **1051** on the first substrate **101** overlaps with the orthographic projections of a first trunk electrode **1062** and a data line **1043** on the first substrate **101** without overlapping with the orthographic projection of the branch electrode **1064** on the first substrate **101**. The orthographic projection of the second transparent common electrode line **1052** on the first substrate **101** overlaps with the orthographic projections of the plurality of branch electrodes **1064**, another data line **1043**, and the second trunk electrode **1063** on the first substrate **101** without overlapping with the first trunk electrode **1062**. With this arrangement, the first transparent common electrode line **1051** and the second transparent common electrode line **1052** both function as a shielding electrode. The first transparent common electrode line **1051** can ameliorate the problem of dark fringes around the first trunk electrode **1062**. The storage capacitor formed between the second transparent common electrode line **1052** and the pixel electrode **1061** is larger, and the problem caused by electric leakage of the storage capacitor can be ameliorated.

(66) In some other embodiments, the orthographic projection of a first transparent common electrode line **1051** on the first substrate **101** may also overlap with the orthographic projections of a first trunk electrode **1062**, a plurality of branch electrodes **1064**, and a data line **1043** on the first substrate **101**.

(67) In some other embodiments, the orthographic projection of the gap between the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** overlaps with the orthographic projection of the branch electrode **1064** on the first substrate **101**. The area where the first transparent common electrode line **1051** overlaps with the second transparent common electrode line **1052** and the pixel electrode **1061** is increased.

(68) In some other embodiments, the minimum pitch d in the second direction y between the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other in the second direction y is greater than or equal to 2.5 microns and less than or equal to 3.5 microns. With this arrangement, the area in which the first transparent common electrode line **1051** overlaps with the second transparent common electrode line **1052** and the pixel electrode **1061** is increased, meanwhile the process accuracy of forming the first transparent common electrode line **1051** and the second transparent common electrode line **1052** is accommodated, thereby simplifying the manufacturing difficulty of the display panel **100**. The minimum pitch d in the second direction y between the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other in the second direction y being too small causes the process of manufacturing the first transparent common electrode line **1051** and the second transparent common electrode line **1052** to be too difficult. The minimum pitch d in the second direction y between the first transparent common electrode line **1051** and the second transparent common electrode line **1052** adjacent to each other in the second direction y being too large causes the utilization of the transparent conductive layer **105** to be

reduced.

(69) In some other embodiments, the orthographic projections of a first transparent common electrode line **1051** and a second transparent common electrode line **1052** adjacent to each other in the second direction y on the first substrate **101** do not overlap with the orthographic projections of the gate **1021** and the active layer **1031** of the thin-film transistor T on the first substrate **101**. With this arrangement, the influences of the first transparent common electrode line **1051** and the second transparent common electrode line **1052** on the gate **1021** and the active layer **1031** are reduced.

(70) Referring to FIG. 5, there is shown a schematic plan view of a pixel electrode layer, a first metal layer, a second metal layer, a semiconductor layer, and a transparent conductive layer according to still some other embodiments of the present disclosure. The pixel electrode **1061** shown in FIG. 5 is substantially similar to the pixel electrodes **1061** shown in FIGS. 3 and 4, and the similarities are not repeated in detail, the differences lie in that the pixel electrode **1061** further includes a third trunk electrode **1067** extending in the first direction x. The third trunk electrode **1067** is arranged parallel to and opposite to the first trunk electrode **1062**, another end of the second trunk electrode **1063** is connected between the two ends of the third trunk electrode **1067**, a portion of the branch electrodes **1064** is connected to the third trunk electrode **1067**, and the second transparent common electrode line **1052** is overlapped with the third trunk electrode **1067**.

(71) In some embodiments of the present disclosure, in the case where the pixel electrode **1061** includes both the third trunk electrode **1067** and the first trunk electrode **1062**, the orthographic projection of the first transparent common electrode line **1051** on the first substrate **101** overlaps with the orthographic projections of the first trunk electrode **1062** and a data line **1043** on the first substrate **101**. Moreover, the orthographic projection of the second transparent common electrode line **1052** on the first substrate **101** overlaps with the orthographic projections of the third trunk electrode **1067**, another data line **1043**, and the plurality of branch electrodes **1064** on the first substrate **101**.

(72) In some other embodiments, the width of the third trunk electrode **1067** is the same as the width of the first trunk electrode **1062**.

(73) Voltages are applied to the first transparent common electrode line **1051** and the second transparent common electrode line **1052**, respectively. The pretilt angle(s) of the liquid crystal molecules **301** under the action of the electric field formed by the voltage difference between the pixel electrode **1061** and the first transparent common electrode line **1051** and the pretilt angle(s) of the liquid crystal molecules **301** under the action of the electric field formed by the voltage difference between the pixel electrode **1061** and the second transparent common electrode line **1052** can be independently controlled. The liquid crystal molecules **301** in the liquid crystal layer **30** are tilted towards outside of the aperture region in the second direction y, thereby ameliorating the problem of the dark fringes in the vicinity of the third trunk electrode **1067** and the first trunk electrode **1062** of the pixel electrode **1061** in the second direction y, which in turn improves the light transmittance of the display panel **100** and improves the display effect of the display device **300**.

(74) Referring to FIG. 6, there is shown a schematic flow diagram of an alignment method of liquid crystal molecules **301** of a display panel of a display device according to some embodiments of the present disclosure. The display panel **100** may be a display panel of the display device **300** of any of the above embodiments, and will not be described herein again. The alignment method of the liquid crystal molecules **301** of the display panel **100** includes steps **S101** and **S102**.

(75) At step **S101**, a first preset alignment voltage is applied to the first transparent common electrode line **1051**.

(76) At step **S102**, a second preset alignment voltage is applied to the second transparent common electrode line **1052**, and the second preset alignment voltage is different from the first preset alignment voltage.

(77) It should be noted that the applying of the first predetermined alignment voltage to the first

transparent common electrode line **1051** and the applying of the second predetermined alignment voltage to the second transparent common electrode line **1052** may be performed simultaneously, but is not limited thereto. In addition, the above method further includes applying a preset alignment common voltage to the common electrode layer **202** without applying a voltage to the plurality of pixel electrodes **1061**.

(78) During the liquid crystal alignment process, the absolute value of the difference between the voltage of the pixel electrode **1061** (0V) and the first preset alignment voltage is different from the absolute value of the difference between the voltage of the pixel electrode **1061** and the second preset alignment voltage. With this arrangement, the intensity of the electric field corresponding to the voltage difference between the first transparent common electrode line **1051** and the pixel electrode **1061** is different from the intensity of the electric field corresponding to the voltage difference between the second transparent common electrode line **1052** and the pixel electrode **1061**. The pretilt angle(s) of the liquid crystal molecules **301** under the action of the electric field corresponding to the voltage difference between the first transparent common electrode line **1051** and the pixel electrode **1061** is different from the pretilt(s) angle of the liquid crystal molecules **301** under the action of the electric field corresponding to the voltage difference between the second transparent common electrode line **1052** and the pixel electrode **1061**.

(79) For example, during liquid crystal alignment, no voltage is applied to the pixel electrode **1061**. The alignment common voltage applied to the common electrode layer **202** is 15V. The first preset alignment voltage applied to the first transparent common electrode line **1051** is -8V. The second preset alignment voltage applied to the second transparent common electrode line **1052** is -3V.

(80) Therefore, the absolute value of the difference between the voltage applied to the pixel electrode **1061** and the first preset alignment voltage applied to the first transparent common electrode line **1051** is 8V. The absolute value of the difference between the voltage applied to the pixel electrode **1061** and the voltage applied to the second transparent common electrode line **1052** is 3 V. The intensity of the electric field formed between the pixel electrode **1061** and the first transparent common electrode line **1051** is stronger than the intensity of the electric field formed between the pixel electrode **1061** and the second transparent common electrode line **1052**. The pretilt angle a of the liquid crystal molecules **301** (for example, the liquid crystal molecules **301A** at the edge of the branch electrode in FIG. 1) overlapped with the first transparent common electrode line **1051** is larger than the pretilt angle b of the liquid crystal molecules **301** (for example, the liquid crystal molecules **301B** at the edge of the branch electrode in FIG. 1) overlapped with the second transparent common electrode line **1052**. In the case where the first transparent common electrode line **1051** overlaps with the first trunk electrode **1062**, the liquid crystal molecules **301** overlapped with the first transparent common electrode line **1051** are more easily to be tilted towards the outside of the aperture region. That is, the liquid crystal molecules **301** on the first trunk electrode **1062** are more easily to be tilted towards a side away from the branch electrode **1064** (connected to the first trunk electrode **1062**), which in turn controls the dark fringes in the vicinity of the first trunk electrode **1062** in the second direction y to be within the non-aperture region, thereby improving the transmittance of the display panel **100** with respect to the backlight and improving the display effect of the display device **300**.

(81) It should be noted that the pretilt angle takes the thickness direction of the display panel as the reference direction, and the larger the angle deviated from the thickness direction, the larger the pretilt angle. For example, the angle at which the liquid crystal molecules **301A** deviate from the reference direction is larger than the angle at which the liquid crystal molecules **301B** deviate from the reference direction. In addition, the display panel **100** further includes two alignment layers (not shown), with one alignment layer being located on the array substrate **10** and disposed on the side of the pixel electrode layer **106** away from the first substrate **101**, and the other alignment layer being located on the opposing substrate **20** and disposed on the side of the common electrode layer **202** away from the second substrate **201**. The pretilt angle(s) of the plurality of liquid crystal

molecules **301** after alignment is fixed by two alignment layers.

(82) It should also be noted that in the case where the pixel electrode **1061** assumes the pixel electrode as shown in FIG. 5, the voltages applied to the first transparent common electrode line **1051** and the second transparent common electrode line **1052** can be independently controlled. The liquid crystal molecules **301** adjacent to the first trunk electrode **1062** and the liquid crystal molecules **301** adjacent to the third trunk electrode **1067** are both tilted towards outside of the aperture region (a side away from the pixel electrode), which in turn controls the dark fringes of the pixel electrode **1061** on both sides in the second direction y to be within the non-aperture region, thereby improving the transmittance of the display panel **100** with respect to the backlight and improving the display effect of the display device **300**.

(83) In some other embodiments, the second preset alignment voltage may also be the same as the first preset alignment voltage.

(84) The above description of the embodiments is merely intended to assist in understanding the technical solution of the present disclosure and the core concepts thereof. It will be appreciated by those of ordinary skill in the art that modifications may still be made to the technical solutions described in the foregoing embodiments, or equivalents may be made to some of the technical features therein. These modifications or substitutions do not depart the essence of the corresponding technical solutions from the scope of the technical solutions of the embodiments of the present disclosure.

Claims

1. An array substrate comprising: a first substrate; a plurality of data lines extending in a first direction and arranged at intervals in a second direction, disposed on the first substrate, the first direction intersecting the second direction; a pixel electrode layer disposed on the first substrate and including a plurality of pixel electrodes, wherein the pixel electrode includes a first trunk electrode extending in the first direction; first transparent common electrode lines extending in the first direction, disposed on the first substrate and insulated from the pixel electrode layer; and second transparent common electrode lines extending in the first direction, disposed on the first substrate and insulated from the pixel electrode layer, and disposed disconnected from the first transparent common electrode lines; wherein an orthographic projection of a gap between one of the first transparent common electrode lines and one of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of at least one of the pixel electrodes on the first substrate, wherein an area where an orthographic projection of a first transparent common electrode line on the first substrate overlaps with an orthographic projection of the first trunk electrode on the first substrate is larger than an area where an orthographic projection of a second transparent common electrode line on the first substrate overlaps with an orthographic projection of the first trunk electrode on the first substrate, wherein an area of the second transparent common electrode line is greater than an area of the first transparent common electrode line.
2. The array substrate of claim 1, wherein a plurality of the first transparent common electrode lines and a plurality of the second transparent common electrode lines are alternately arranged in the second direction.
3. The array substrate of claim 2, wherein the array substrate further comprising: a first trunk common electrode line extending in the second direction, connected to a plurality of the first transparent common electrode lines; and a second trunk common electrode line extending in the second direction, disposed disconnected from the first trunk common electrode line and connected to a plurality of the second transparent common electrode lines.
4. The array substrate of claim 1, wherein an orthographic projection of at least one of a first transparent common electrode line of the first transparent common electrode lines and a second

transparent common electrode line of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of at least one of the pixel electrodes on the first substrate.

5. The array substrate of claim 1, wherein the pixel electrode includes a plurality of branch electrodes arranged at intervals and a plurality of apertures arranged at intervals, one of the apertures being disposed between two adjacent ones of the branch electrodes; an orthographic projection of at least one of a first transparent common electrode line of the first transparent common electrode lines and a second transparent common electrode line of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of the branch electrode of the at least one of the pixel electrode on the first substrate.

6. The array substrate of claim 4, wherein the pixel electrode includes a plurality of branch electrodes arranged at intervals and a plurality of apertures arranged at intervals, one of the apertures being disposed between two adjacent ones of the branch electrodes; an orthographic projection of at least one of a first transparent common electrode line of the first transparent common electrode lines and a second transparent common electrode line of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of the branch electrode of the at least one of the pixel electrode on the first substrate.

7. The array substrate of claim 1, wherein the array substrate comprising a transparent conductive layer including the first transparent common electrode lines and the second transparent common electrode lines; wherein the pixel electrode layer is located on a side, away from the first substrate, of a film layer on which the plurality of data lines are located, and the transparent conductive layer is located between the film layer on which the plurality of data lines are located and the pixel electrode layer.

8. The array substrate of claim 4, wherein the array substrate comprising a transparent conductive layer including the first transparent common electrode lines and the second transparent common electrode lines; wherein the pixel electrode layer is located on a side, away from the first substrate, of a film layer on which the plurality of data lines are located, and the transparent conductive layer is located between the film layer on which the plurality of data lines are located and the pixel electrode layer.

9. The array substrate of claim 7, wherein an orthographic projection of at least one of a first transparent common electrode line of the first transparent common electrode lines and a second transparent common electrode line of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of at least one of the data lines on the first substrate.

10. The array substrate of claim 8, wherein an orthographic projection of at least one of a first transparent common electrode line of the first transparent common electrode lines and a second transparent common electrode line of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of at least one of the data lines on the first substrate.

11. The array substrate of claim 7, wherein the array substrate further comprising: a color resistance layer located between a film layer on which the plurality of data lines are located and the transparent conductive layer.

12. The array substrate of claim 8, wherein the array substrate further comprising: a color resistance layer located between a film layer on which the plurality of data lines are located and the transparent conductive layer.

13. The array substrate of claim 1, wherein the pixel electrode is of an asymmetric structure in the second direction.

14. The array substrate of claim 1, wherein the pixel electrode includes: a second trunk electrode

extending in the second direction, one end of the second trunk electrode being connected between both ends of the first trunk electrode; a plurality of branch electrodes located on both sides of the second trunk electrode in the first direction and respectively connected to the first trunk electrode and/or the second trunk electrode; wherein an orthographic projection of one of the first transparent common electrode lines on the first substrate overlaps with orthographic projections of the first trunk electrode and one of the data lines on the first substrate, and an orthographic projection of one of the second transparent common electrode lines on the first substrate overlaps with orthographic projections of a portion of the branch electrodes and another one of the data lines on the first substrate.

15. The array substrate of claim 13, wherein the pixel electrode includes: a second trunk electrode extending in the second direction, one end of the second trunk electrode being connected between both ends of the first trunk electrode; a plurality of branch electrodes located on both sides of the second trunk electrode in the first direction and respectively connected to the first trunk electrode and/or the second trunk electrode; wherein an orthographic projection of one of the first transparent common electrode lines on the first substrate overlaps with orthographic projections of the first trunk electrode and one of the data lines on the first substrate, and an orthographic projection of one of the second transparent common electrode lines on the first substrate overlaps with orthographic projections of a portion of the branch electrodes and another one of the data lines on the first substrate.

16. A display panel comprising an array substrate, an opposing substrate, and a liquid crystal layer, the opposing substrate is disposed opposite to the array substrate, and the liquid crystal layer is disposed between the array substrate and the opposing substrate; the array substrate includes: a first substrate; a plurality of data lines extending in a first direction and arranged at intervals in a second direction, disposed on the first substrate, the first direction intersecting the second direction; a pixel electrode layer disposed on the first substrate and including a plurality of pixel electrodes, wherein the pixel electrode includes a first trunk electrode extending in the first direction; first transparent common electrode lines extending in the first direction, disposed on the first substrate and insulated from the pixel electrode layer; and second transparent common electrode lines extending in the first direction, disposed on the first substrate and insulated from the pixel electrode layer, and disposed disconnected from the first transparent common electrode lines; wherein an orthographic projection of a gap between one of the first transparent common electrode lines and one of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of at least one of the pixel electrodes on the first substrate; wherein orthographic projections of one of the first transparent common electrode lines and one of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate do not overlap with an orthographic projection of a gate of a thin film transistor on the first substrate, wherein an area of the second transparent common electrode line is greater than an area of the first transparent common electrode line.

17. The display panel of claim 16, wherein a plurality of the first transparent common electrode lines and a plurality of the second transparent common electrode lines are alternately arranged in the second direction.

18. The display panel of claim 17, wherein the array substrate further comprising: a first trunk common electrode line extending in the second direction, connected to a plurality of the first transparent common electrode lines; and a second trunk common electrode line extending in the second direction, disposed disconnected from the first trunk common electrode line and connected to a plurality of the second transparent common electrode lines.

19. The display panel of claim 16, wherein an orthographic projection of at least one of a first transparent common electrode line of the first transparent common electrode lines and a second transparent common electrode line of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of

at least one of the pixel electrodes on the first substrate.

20. An alignment method of liquid crystal molecules, applied to a display panel, wherein the display panel comprises an array substrate and an opposing substrate disposed opposite each other, and a liquid crystal layer disposed between the array substrate and the opposing substrate and including a plurality of liquid crystal molecules; the array substrate includes a first substrate and a pixel electrode layer disposed on the first substrate, first transparent common electrode lines extending in a first direction, second transparent common electrode lines extending in the first direction, and a plurality of data lines extending in a first direction and arranged at intervals in a second direction; the pixel electrode layer including a plurality of pixel electrodes, wherein the pixel electrode includes a first trunk electrode extending in the first direction, the first transparent common electrode lines and the second transparent common electrode lines disposed disconnected from each other and insulated from the pixel electrode layer; an orthographic projection of a gap between one of the first transparent common electrode lines and one of the second transparent common electrode lines adjacent to each other in the second direction on the first substrate overlaps with an orthographic projection of at least one of the pixel electrodes on the first substrate, wherein an area where an orthographic projection of a first transparent common electrode line on the first substrate overlaps with an orthographic projection of the first trunk electrode on the first substrate is larger than an area where an orthographic projection of a second transparent common electrode line on the first substrate overlaps with an orthographic projection of the first trunk electrode on the first substrate, wherein an area of the second transparent common electrode line is greater than an area of the first transparent common electrode line, the first direction intersecting the second direction; the method comprising: applying a first preset alignment voltage to the first transparent common electrode lines; applying a second preset alignment voltage to the second transparent common electrode lines, the second preset alignment voltage being different from the first preset alignment voltage.
